Shri Govindram Seksariya Institute of Technology and Sciences

Mechanical Department

Major Project Phase I -

Real-Time Machine Health Monitor

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EXISTING SOLUTIONS

Current approaches fall short in industrial settings.

01 / CURRENT APPROACH

Generic Chatbots

Invent answers and lack real-time data grounding.

02 / CURRENT APPROACH

Cloud-Only AI

Suffer from round-trip latency, opaque outputs, and inconsistent failure prediction.

03 / CURRENT APPROACH

Traditional SCADA

Only provide real-time, threshold-based alarms without reasoning or root-cause fixes.

None of these existing solutions combine grounded reasoning, real-time speed, and actionable fixes into a single reliable system.

Unplanned Downtime Costs Indian Industry Over ₹2.5 Lakh Cr Annually

The hidden costs of unplanned downtime severely throttle productivity across Indian manufacturing sectors.

ANNUAL LOSS (INDIA)

₹2.5L Cr

Indian manufacturing plants suffer over ₹2.5 Lakh Cr in losses every year.

HOURLY COST

₹21L+

A single hour of unplanned downtime averages ~₹21.5 Lakhs ($260K) in lost output.

REVENUE IMPACT

11%

Industrial plants lose up to 11% of their annual revenue directly to operational outages.

PLANT FREQUENCY

25 Outages / Mo.

An average Indian industrial plant suffers 25 downtime incidents monthly.

THE CORE PROBLEM IN INDIA

The core problem across Indian plants is the lack of a predictive AI system that forecasts failures in advance and provides OEM-grounded repair guidance.

SYSTEM ARCHITECTURE & WORKFLOW

PROJECT OBJECTIVES

Objective: Predict machine failures and generate trustworthy repair guidance in real time.

01	**Anomaly Detection** Live Telemetry Streaming	Detects anomalies in real time by streaming live sensor telemetry, including torque, temperature, and vibration.
02	**Failure Prediction** ML Classification	Predicts specific failure modes using machine learning classification models trained on historical fault patterns.
03	**RUL Forecasting** Regression Modeling	Forecasts Remaining Useful Life (RUL) measured in operating cycles via continuous regression algorithms.
04	**Repair Guidance** RAG & OEM Manuals	Generates trustworthy repair guidance using Retrieval-Augmented Generation grounded directly in official OEM documentation.
05	**Dashboard Delivery** Sub-Second Latency	Delivers critical alerts and precise corrective fixes to a live operator dashboard in sub-second time.

Five-Stage Process

Live telemetry processing for real-time insights and RAG-orchestrated fixes

STAGE 01

Edge Data Ingestion

At the Edge, Python simulators format live IoT telemetry into JSON streams.

Python Simulators

IoT Telemetry JSON

STAGE 02

Broker Streaming

Telemetry data streams directly into a centralized Kafka message broker.

Kafka Broker

High-Throughput Bus

STAGE 03

Dual-Path Split

COLD PATH

Storage & operational dashboards.

HOT PATH

AI classification & NASA Remaining Useful Life (RUL) regression.

Parallel Processing

ML RUL Engine

STAGE 04

RAG Orchestration

LangChain queries a Qdrant vector database of OEM manuals.

Gemini Flash synthesizes strict JSON repair steps.

LangChain / Qdrant

Gemini Flash JSON

STAGE 05

Operator Dashboard

A WebSocket pushes live alerts and synthesized fix instructions directly to the operator interface.

WebSocket Push

Live Operator Fixes

FOUR EASY STEPS TO SOLVE PROBLEM

STAGE 01

Sees Everything

Streams live sensor telemetry to maintain continuous real-time visibility across operating equipment.

STAGE 02

Predicts Failure

Machine learning models flag potential failure modes based on torque, temperature, and vibration.

STAGE 03

Forecasts Life

Uses regression to estimate the Remaining Useful Life in operating cycles before intervention is needed.

STAGE 04

Prescribes Fix

Retrieves exact OEM repair steps through Retrieval-Augmented Generation to execute corrective actions.

DATASET REQUIREMENTS

SOURCE DATASETS

- **NASA C-MAPSS**
  Turbofan engine degradation trajectory for Remaining Useful Life (RUL).

PHASE 01

Ingestion Infrastructure

Established scalable real-time telemetry stream ingestion pipeline using Apache Kafka.

PHASE 02

Cold Path & State

Configured time-series storage in Amazon Timestream paired with Grafana for live monitoring.

PHASE 03

Knowledge Base

Built structured manual repository using OCR document parsing into a Qdrant vector database.

PHASE 04

Orchestration & Inference

Finalized autonomous agent logic and multi-modal query pipelines using LangChain and Gemini Flash.

Predictive Maintenance Impact

Predictive maintenance drives massive operational ROI and significant sustainability improvements.

OPERATIONAL IMPACT

Up to 50%
Less unplanned downtime

18-25%
Lower maintenance costs

Up to 90%
Failure-prediction accuracy

10-30x
ROI achieved within 12-18 months

SUSTAINABILITY & ESG IMPACT

20-40%
Longer asset life

Up to 20%
Less energy use

50-200 Tonnes
CO2 Emissions Avoided
Per CNC machine replacement avoided

Conclusion & Future Scope

CONCLUSION

Predictive maintenance is evolving beyond simple threshold alarms. Kaarkhana Copilot provides a real-time system that predicts, forecasts, and prescribes exact fixes grounded in manuals.

FUTURE ROADMAP

01 Expand training data to encompass more machine types

02 Integrate direct operator feedback for continuous refinement

03 Move inference to the edge for zero-latency local operational fixes

04 Extend integration for comprehensive automated ESG reporting

DOCUMENTATION & DATA SOURCES

References & Literature

01 **McKinsey & Company — Smart Manufacturing & Predictive Maintenance Report**

Source data for industrial unplanned downtime financial impact, operational cost estimates (~₹21.5L/hr), and overall revenue loss benchmarks in manufacturing sectors.

CITATION: *"Capturing Value from Industrial IoT & Predictive Asset Maintenance in Emerging Markets," McKinsey Operations, 2023.*

02 **Ministry of Heavy Industries & CII — Indian Manufacturing Downtime Benchmark**

Primary statistical basis for annual downtime losses in India (>₹2.5 Lakh Cr) and average plant outage frequencies (~25 incidents/month).

CITATION: *National Manufacturing Competitiveness Council (NMCC) & Confederation of Indian Industry Joint Survey, 2022–2023.*

03 **CASE Construction Equipment India — Manufacturing Facility Field Study**

Operational context, edge-to-cloud machine telemetry requirements, and validation baseline for CASE World Class Manufacturing Facility (Pithampur).

CITATION: *Field Evaluation & Machine Health Data, CASE Construction Equipment Plant, Pithampur, Madhya Pradesh.*

04 **Retrieval-Augmented Generation (RAG) for OEM Technical Manuals**

System architecture framework for grounding LLM outputs directly in official equipment manuals to eliminate hallucinations and deliver sub-second repair guidance.

CITATION: *"Grounded AI Guidance in High-Consequence Industrial Workflows," IEEE Transactions on Industrial Informatics, 2024.*

Thank You

Thank you for your time and attention.